BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Truth Tables and Combined Logic

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How can you recognise a NOT gate symbol?

2. What does an encryption utility produce?

A) A smaller file
B) Ciphertext
C) A sorted folder
D) Extra storage space

3. "Publicly posting all personal records always improves privacy." Correct this. Explain why.

4. A login must get details, check them and choose the next step. "Algorithmic thinking means guessing steps in a random order." Correct this. Why does order matter?

Today's objectives

Explain the core ideas of truth tables and combined logic using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Combined logic
Work out intermediate gate outputs before calculating the final result.

Operator precedence
Use brackets to clarify order when translating logic diagrams.

Three inputs
A truth table has $2^3 = 8$ unique input combinations.

Intermediate columns
Recording a gate's result reduces mistakes in multi-stage circuits.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Combined logic	Work out intermediate gate outputs before calculating the final result.
Operator precedence	Use brackets to clarify order when translating logic diagrams.
Three inputs	A truth table has $2^3 = 8$ unique input combinations.
Intermediate columns	Recording a gate's result reduces mistakes in multi-stage circuits.

I Do - One worked example

Complete Q = (A AND B) OR C for all eight input combinations.

1. List 000,001,010,011,100,101,110,111 in ABC order.
2. Find X=A AND B: 0,0,0,0,0,0,1,1.
3. Then Q=X OR C: 0,1,0,1,0,1,1,1.
4. Check rows with C=1 must have Q=1.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Find Q=(A OR B) AND C when A=1, B=0, C=0 and then C=1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition so A with B, or C alone, raises an alert.

```
A = true
B = false
C = true
if ... then
print("Investigate")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. How many rows are needed for a 3-input truth table? [2 marks]

Q3. Evaluate (A AND B) OR C when A=1 B=0 C=1. [2 marks]

Q4. Explain the benefit of an intermediate X column. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A three-input truth table has only four rows. ☐ T ☐ F
2. Work out intermediate gate outputs before calculating the final result. ☐ T ☐ F
3. Use brackets to clarify order when translating logic diagrams. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It is a one-input inverter, usually a triangle with a small output bubble. ☐

Q2 B - Ciphertext ☐

Q3 People may lose control over how identifying information is used or shared. ☐

Q4 Plan clear steps in the right order to solve a problem. Keep needed details, split the problem into parts, then link the steps. ☐

Code First - Model

```
A = true
B = false
C = true
if (A AND B) OR C then
print("Investigate")
endif
```

Output: Investigate

We Do - Check the model

1. Explain why AND is applied last. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 $2^3=8$. ☐

Q3 1. ☐

Q4 Allows accurate separate evaluation of one gate and helps trace errors in combined expressions. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: